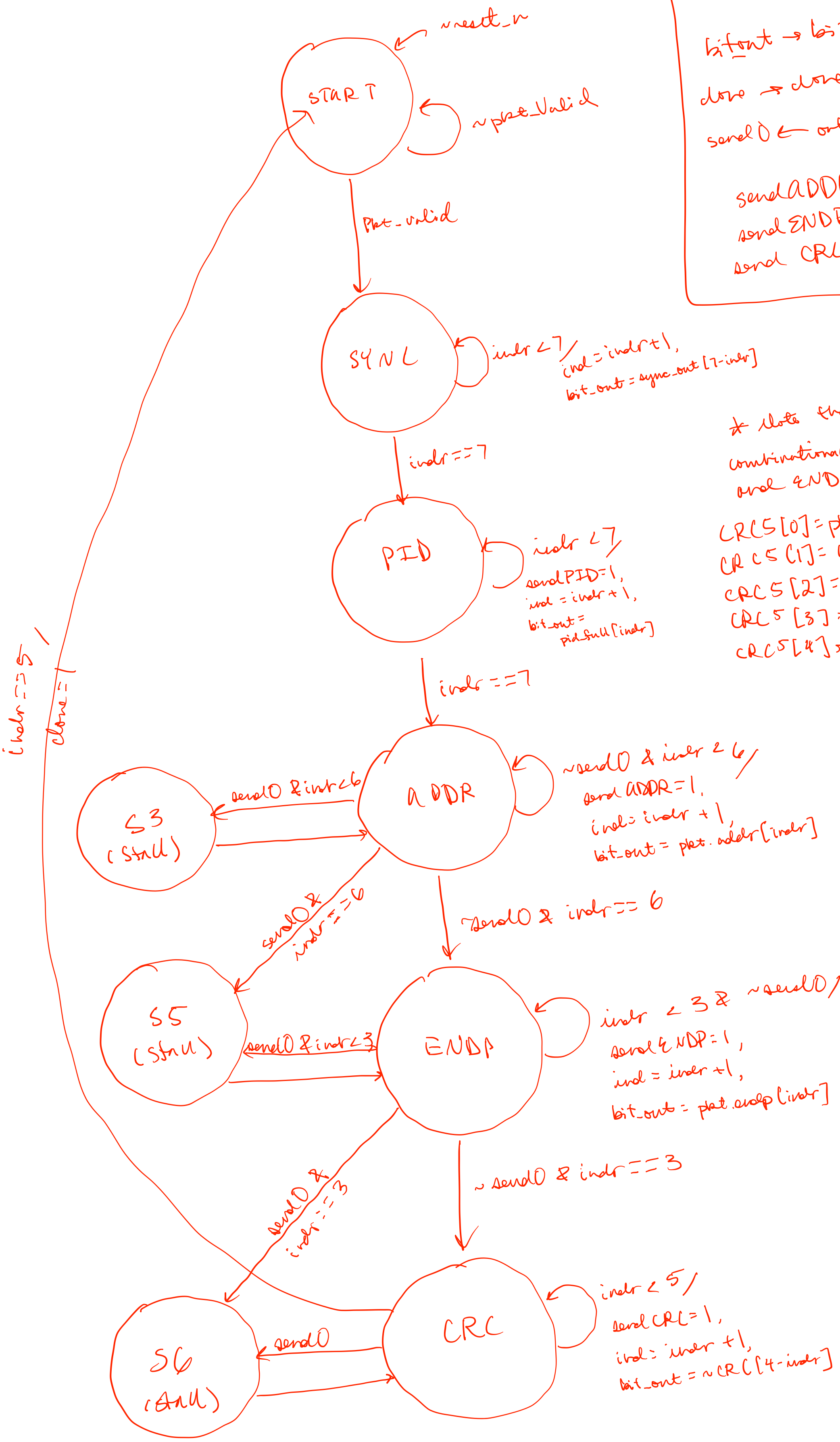
CRC5 FSM



Handshake signals:

bitout → bit_in input for BS FSM

done → done input for NRZI FSM

send0 ← output from BS

sendADDR → sendAddr input for BS

send ENDP → sendENDP input for BS

send CRC → sendCRC input for BS

* Note that the CRC is computed combinationally each cycle of ADDR and ENDP where:

CRC5[0] = pkt.addr[indr] ^ CRC_r[4]

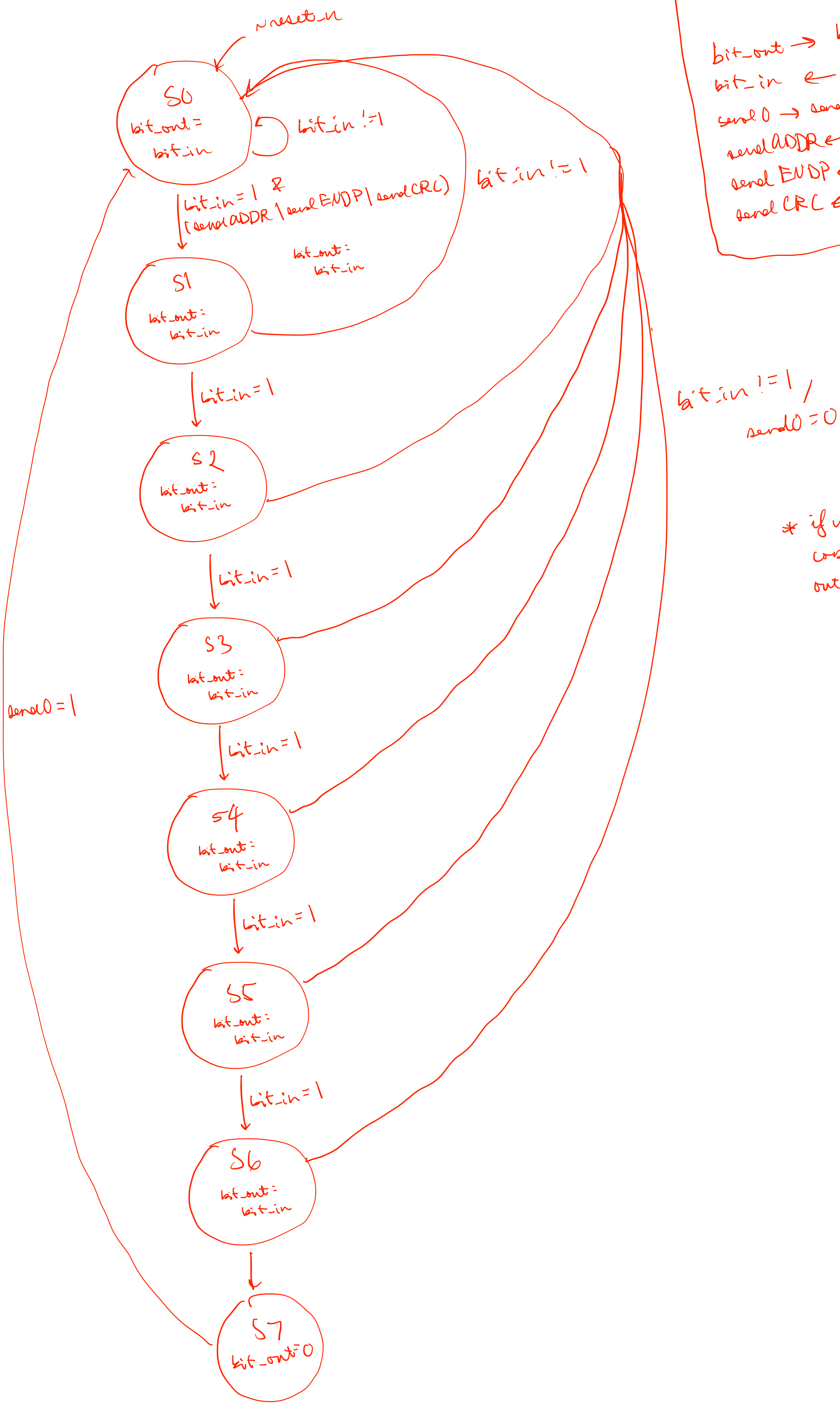
CRC5[1] = CRC_r[0]

CRC5[2] = (pkt.addr[indr] ^ CRC_r[4]) ^ CRC_r[1]

CRC5[3] = CRC_r[2]

CRC5[4] = CRC_r[3]

BS FSM



handshake signals:

bit_out → bit_in input for NRZI
bit_in ← bitout output from CRC
send0 → send0 input for CRC
sendADDR ← sendADDR output from CRC
sendENDP ← sendENDP output from CRC
sendCRC ← sendCRC output from CRC

* if we receive 6 consecutive 1s, send out a 0

NRZI FSM

handshake signals:

done ← done output from CRC

bit_in ← bit_out output from BS